

A PROJECT REPORT ON

MediAI : AI-ML Based Disease Detection System

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

SUBMITTED BY

SARTHAK DHAYTONDE	Exam No: B400050376
AJIT KALE	Exam No: B400050432
KARTIK KASREWAR	Exam No: B400050440
SOHAM LABBA	Exam No: B400050467



DEPARTMENT OF COMPUTER ENGINEERING
Pune Institute of Computer Technology
Dhankawadi, Pune - 411043

SAVITRIBAI PHULE PUNE UNIVERSITY
2024-2025



CERTIFICATE

This is to certify that the project report entitled
MediAI : AI-ML Based Disease Detection System

Submitted by

Sarthak Dhaytonde

Exam No: B400050376

Ajit Kale

Exam No: B400050432

Kartik Kasrewar

Exam No: B400050440

Soham Labba

Exam No: B400050467

are bonafide students of this institute and the work has been carried out by them under the supervision of **Prof. Y.A. Handge** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of **Bachelor of Engineering** (Computer Engineering).

Prof. Y.A. Handge

Internal Guide

Dept. of Computer Engg.

Dr. G. V. Kale

Head

Dept. of Computer Engg.

Dr. S. T. Gandhe

Principal

Pune Institute of Computer Technology

Place: Pune

Date: 16/05/2025

ACKNOWLEDGEMENT

*It gives us great pleasure in presenting the project report on ‘**MediAI : AI-ML Based Disease Detection System**’.*

*We would like to thank our project guide **Prof. Y.A. Handge** for giving us all the help and guidance we needed. We are really grateful to them for their kind support and valuable suggestions.*

*We are also grateful to **Dr G. V. Kale**, Head of Computer Engineering Department, PICT for her indispensable support and suggestions throughout the project. We also express our gratitude to **Dr. A. R. Deshpande**, BE project coordinator, for her timely guidance and support.*

SARTHAK DHAYTONDE
AJIT KALE
KARTIK KASREWAR
SOHAM LABBA
(B.E. Computer Engg.)

ABSTRACT

The MediAI: Disease Detection System is a web-based healthcare platform that leverages artificial intelligence and machine learning to assist users in early disease detection based on their symptoms. It aims to bridge the gap between patients and healthcare providers by offering a reliable, data-driven tool that predicts potential illnesses and recommends relevant specialists. Users interact with a symptom checker powered by AI algorithms trained on diverse medical datasets, enabling real-time, accurate predictions. The platform is designed with a focus on accessibility, offering a seamless user experience across devices, and features a generative chatbot that answers queries related to health concerns using natural language processing techniques.

Beyond prediction, MediAI introduces a secure, centralized medical records system, allowing patients to upload their medical history and share it effortlessly with doctors via QR code scans. This feature enhances continuity of care, enabling physicians to make more informed decisions with direct access to critical health data. The platform is built on a modern technology stack, including ReactJS, Node.js, Python-based AI models, and MongoDB for scalable and responsive functionality. Security is prioritized through robust encryption protocols and user authentication,

This project addresses pressing issues in global and rural healthcare accessibility, misdiagnosis, and inefficient patient-provider communication. It includes a detailed development methodology, covering AI model integration, frontend/backend engineering, and user interface design, along with a comprehensive project timeline and risk management strategy. Ultimately, MediAI seeks to transform digital health services by combining intelligent automation, secure data handling, and user-friendly interaction—laying the groundwork for scalable, AI-powered healthcare solutions that prioritize patient outcomes, transparency, and trust.

Contents

1	Introduction	1
1.1	Overview	2
1.2	Motivation	2
1.3	Problem Definition and Objectives	2
1.3.1	Problem Definition	2
1.3.2	Objectives	3
1.4	Project Scope & Limitations	3
1.4.1	Limitations	3
2	Literature Survey	5
3	Software Requirements Specification	9
3.1	Assumptions and Dependencies	10
3.1.1	Assumptions	10
3.1.2	Dependencies	10
3.2	Functional Requirements	11
3.3	Non-Functional Requirements	11
3.4	External Interface Requirements	13
3.4.1	Hardware Interfaces	13
3.4.2	Software Interfaces	13
4	System Design	14
4.1	Model Architecture	15
4.2	UML Diagrams	16
4.2.1	Usecase Diagram	16
5	Project Plan	19
5.1	Risk Management	20
5.1.1	Risk Identification	20
5.1.2	Risk Analysis	20
5.1.3	Overview of Risk Mitigation, Monitoring, Management	20
5.2	Project Schedule	21
5.2.1	Project Task Set	21
5.2.2	Team Structure	21
6	Implementation	22
6.1	Phases of Execution	23
6.1.1	Data Preparation	23
6.1.2	Execution of Modules	23
6.2	Experimental Configuration	24

6.2.1	Specification of Experiment Scenarios	24
6.2.2	Tools and Software Used	25
7	Results	26
8	Conclusion	29
8.1	Conclusion	30
9	Appendix	31
9.1	Appendix A	32
9.1.1	Technical Feasibility	32
9.1.2	Operational Feasibility	32
9.1.3	Economic Feasibility	32
9.2	Appendix B	33
9.2.1	Survey Paper Details	33
9.2.2	Implementation Paper Details	34
9.3	Appendix C	35
9.3.1	Plagiarism Report	35
10	References	36

List of Figures

4.1	Model Architecture	15
4.2	Use case diagram	16
4.3	State Diagram	17
4.4	Class Diagram	18
7.1	Homepage	27
7.2	Uploading Documents	27
7.3	Selection of Symptoms	28
7.4	Result with Bar Chart	28
9.1	Plagarism Report	35

CHAPTER 1

INTRODUCTION

1.1 Overview

MediAI is an intelligent healthcare platform designed to enhance early disease detection and streamline patient-doctor interactions using machine learning, natural language processing, and secure cloud technologies. The system enables users to input symptoms through an intuitive web interface, where trained AI models predict the most likely illnesses and recommend suitable specialists. It further empowers healthcare delivery by allowing patients to upload medical records, which are securely stored and accessible by registered doctors through QR-code-based retrieval. Built using modern frameworks like ReactJS, Node.js, and Python for AI services, MediAI also integrates a chatbot powered by NLP to provide reliable, real-time answers to health-related queries. With a user-focused design and emphasis on data security, the platform bridges the gap between accessible health insights and professional medical care.

1.2 Motivation

Traditional healthcare systems often suffer from slow diagnostic turnaround, fragmented medical records, and limited accessibility—especially in underserved or rural regions. Patients frequently delay treatment due to the lack of immediate medical feedback, while doctors struggle with incomplete patient histories and inefficient data sharing practices. MediAI was conceived to address these systemic challenges by leveraging artificial intelligence for early symptom-based disease prediction and providing a seamless, centralized portal for health record management. The integration of AI-driven diagnostics with secure record sharing offers a holistic approach to digital healthcare, making preventive and primary care more accessible, efficient, and responsive.

1.3 Problem Definition and Objectives

1.3.1 Problem Definition

In many parts of the world, including developing regions, individuals often face barriers to timely and accurate medical diagnosis due to a shortage of healthcare professionals, lack of digitized records, and poor symptom awareness. Patients typically rely on in-person consultations, which are not only time-consuming but often inaccessible for those in remote or underserved areas. Even when care is available, doctors may lack prior medical data needed to make informed decisions. Additionally, there is a growing need for systems that can pre-screen symptoms, guide patients to the right specialists, and support clinical decisions with intelligent recommendations—all while preserving data privacy and usability.

1.3.2 Objectives

- Symptom-Based Disease Prediction: Develop a machine learning model that predicts potential diseases based on user-reported symptoms, using a rich medical dataset and validated classification techniques.
- Medical Record Management: Enable patients to securely upload, store, and manage their medical reports and health history in a cloud-based system, accessible only by authorized professionals.
- QR Code-Based Record Sharing: Generate dynamic QR codes linked to patient records, allowing doctors to retrieve relevant health data securely during consultations. -Powered Chatbot: Integrate a natural language processing (NLP) chatbot capable of answering general health questions, improving user engagement and awareness.
- Security and Privacy Assurance: Implement role-based access control, encryption protocols, and compliance measures (HIPAA/GDPR) to safeguard sensitive health data and ensure ethical handling.
- User-Centric Web Platform: Design an intuitive, responsive web interface accessible to users across various devices and skill levels, promoting widespread adoption.

1.4 Project Scope & Limitations

The platform is intended to assist users in early disease identification by analyzing user-input symptoms using machine learning algorithms. Provides a secure, centralized medical record system allowing patients to upload and manage health data. Facilitates QR-based sharing of medical records with registered doctors, ensuring timely access during consultations. Includes a natural language chatbot for health-related queries and basic awareness. Designed as a responsive, cross-platform web application accessible from desktops and mobile devices. Focuses on individual patient use cases with optional doctor collaboration features; can be extended to clinic-based or telemedicine platforms in future iterations.

1.4.1 Limitations

- Disease prediction is limited by the scope and diversity of the training dataset and may not cover rare or emerging conditions.
- The chatbot cannot provide emergency or legally binding medical advice and serves only for general information.
- The QR code access system assumes the doctor has internet access and device compatibility.

MediAI : AI-ML Based Disease Detection System

- Users must manually enter symptoms; voice-based or wearable integration is not included in the current version.
- The platform does not currently support integration with hospital information systems or EHR standards like HL7/FHIR.

CHAPTER 2

LITERATURE SURVEY

The significance of studying user experience (UX) and AI-driven healthcare systems has gained increasing attention in the healthcare industry, especially as digital health technologies become more complex and accessible. The literature reveals that a well-designed UX not only enhances patient engagement but also plays a crucial role in ensuring the effectiveness of AI-powered predictions and the seamless interaction between patients and healthcare providers. Research indicates that users' perceptions of accuracy and trust in healthcare systems are heavily influenced by the intuitiveness of the platform's interface, the clarity of the AI-driven predictions, and the ease of accessing and sharing medical records. In healthcare platforms, where critical decisions are made based on system outputs, it becomes essential to design interfaces that are user-friendly, transparent, and informative. Studies have shown that healthcare systems with clear, well-communicated AI predictions and robust data security measures lead to greater patient satisfaction and trust. Understanding the interplay between UX and AI-driven healthcare tools is key to identifying the factors that contribute to a successful digital health platform, such as AI model transparency, patient data accessibility, and real-time feedback.

- Alanazi, R. (2022) [50] in their paper "Identification and Prediction of Chronic Diseases Using Machine Learning Approach" explored the use of machine learning techniques to predict common chronic illnesses by analyzing both structured and unstructured data. The study utilized machine learning algorithms, including Convolutional Neural Networks (CNN) for automatic feature extraction and K-nearest neighbor (KNN) for precise distance calculations, to provide accurate disease predictions. The dataset comprised various symptoms, patient living habits, and doctor consultation details, with a focus on predicting diseases like diabetes, cardiovascular disease, and cancer. The paper highlights the importance of leveraging AI in health care to improve early diagnosis and treatment planning, ultimately reducing the mortality rates associated with chronic diseases. The authors further conducted a comparative study, demonstrating that their proposed model outperforms traditional algorithms such as Naive Bayes, decision trees, and logistic regression. They emphasize that incorporating machine learning into healthcare systems can lead to more accurate predictions and better patient outcomes [1].
- Lu, H., Uddin, S. (2023) [50] in their paper titled "Unsupervised Machine Learning for Disease Prediction: A Comparative Performance Analysis Using Multiple Datasets" explored the effectiveness of unsupervised machine learning (ML) techniques in predicting various diseases by utilizing multiple datasets. The study examined the performance of seven unsupervised ML algorithms across 15 datasets, including heart failure, diabetes, and breast cancer, using six performance metrics such as Adjusted Rand Index, Homogeneity, and Silhouette Coefficient. The study revealed that DBSCAN (Density-Based Spatial Clustering of Applications with Noise) performed the best in most cases, followed by Bayesian Gaussian Mixture and Divisive Clustering. The results emphasize that no single model is consistently superior

across all datasets, and model selection should depend on the specific application and performance measures. The authors recommend further research into unsupervised ML methods using real disease records to enhance model selection and disease prediction capabilities.[2].

- KGaurav, AKumar, PSingh, AKumari, MKasar, TSuryawanshi(2023) [50] in their paper titled "Human Disease Prediction using Machine Learning Techniques and Real-Life Parameters" explored the application of machine learning techniques to predict human diseases based on patient symptoms. The study utilized various machine learning algorithms, including Random Forest, Long Short-Term Memory (LSTM), and Support Vector Machine (SVM), to predict diseases such as diabetes, cardiovascular conditions, and others, using a medical dataset from Kaggle. The dataset was transformed by assigning weights to rare symptoms based on geographical locations to improve accuracy. The proposed model showed better accuracy and reliability in comparison to state-of-the-art methods, achieving 97% accuracy. The authors emphasized that this model contributes significantly to automating healthcare diagnosis and improving patient outcomes, making it a useful tool for early disease detection[3].
- Reddy, P.P., Babu, D.M., Kumar, H., Sharma, S. (2021) [50] in their paper titled "Disease Prediction using Machine Learning" explored the use of machine learning techniques, specifically the Random Forest classifier, to predict diseases based on user-reported symptoms. The study emphasizes the importance of an automated system that can predict disease likelihood, thus saving time and effort for patients, especially in regions with limited access to healthcare. The proposed method focuses on analyzing user-input symptoms and predicting the associated disease with a high level of accuracy. The study utilized datasets from various health-related websites and achieved notable accuracy for conditions such as diabetes (98.25%) and kidney disease (99%). This system aims to improve healthcare accessibility and diagnostic accuracy [4].
- Ahmad, A.A., Polat, H. (2023) [50] in their paper titled "Prediction of Heart Disease Based on Machine Learning Using Jellyfish Optimization Algorithm" investigated the application of machine learning (ML) techniques in predicting heart disease, utilizing the Cleveland heart disease dataset. The study focused on selecting the most effective features from the dataset using the Jellyfish optimization algorithm, known for its high convergence speed and flexibility in feature selection, to prevent overfitting due to the high dimensionality of the data. Various ML algorithms were then trained on the optimized dataset, and their performances were evaluated using metrics such as Sensitivity, Specificity, Accuracy, and Area Under Curve (AUC). The support vector machine (SVM) model, trained on the feature-selected dataset, yielded the best results, achieving a Sensitivity of 98.56% and Specificity of 98.37%. The study concludes that the combination of the Jellyfish optimization algorithm with the SVM classifier demonstrates superior performance for heart disease prediction.

The authors recommend further exploration of this approach to improve early detection and predictive accuracy for heart disease[5].

- Alam, M., Fatima, M., Azeem, M. (2020) [50] in their paper “AI based Prediction of Cardiovascular Disease Risk Using Machine Learning Algorithms” explored machine learning algorithms to predict cardiovascular diseases using patient data. They applied models such as Decision Trees, Random Forest, and Support Vector Machine (SVM) on a large cardiovascular dataset, focusing on risk factors like cholesterol levels, blood pressure, and age. The study found that Random Forest outperformed the other models, achieving an accuracy of 94% improve early detection and preventative care for cardiovascular conditions[6].

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- **Reliable Medical Data:** It is assumed that the medical records and symptom data provided by users and third-party sources are accurate, complete, and correctly formatted for AI processing.
- **Accurate Symptom Input:** Users are expected to input their symptoms precisely and truthfully, as the accuracy of disease prediction depends heavily on the quality of user-provided information.
- **Doctor Qualification and Availability:** The system assumes that the listed doctors are qualified, appropriately categorized by specialty, and available for consultation as per the platform's recommendations.
- **Data Privacy Compliance:** It is assumed that all users (patients and doctors) comply with data privacy and consent policies before uploading or accessing medical records through the system.
- **Stable System Environment:** The system is assumed to be deployed on a stable network with minimal downtime and is supported by adequate computing resources to handle AI inference and record storage reliably.

3.1.2 Dependencies

- **Python Libraries:** The MediAI platform relies on core Python libraries such as Flask (for web integration), scikit-learn (for machine learning algorithms), pandas and NumPy (for data manipulation), joblib (for model serialization), and NLTK (for sentiment analysis in chatbot functionality).
- **Machine Learning and AI Models:** Disease prediction is performed using AI models trained on symptom and disease datasets. These models can include classifiers like Decision Trees, Random Forests, or advanced architectures such as BiLSTM, and may be updated or replaced as needed.
- **Web Framework:** The user interface is developed using the Flask microframework, which requires a compatible Python runtime environment and a server setup for local or cloud deployment.
- **Model and Configuration Files:** Serialized model files (e.g., model.pkl), tokenizer configurations, and auxiliary mappings (such as symptom-to-disease mappings) are necessary for accurate predictions and chatbot responses.

- **Database and Storage:** The system depends on a secure server-side database to store and retrieve user medical records, which are accessible via a QR code system by authenticated doctors.
- **Browser Access:** End users must have access to a modern, standards-compliant web browser to interact with the platform, use the chatbot, and upload or view medical records.

3.2 Functional Requirements

1. User Account Management:

- Users must be able to register and log in securely.
- Authentication should support secure password handling.

2. Symptom Checker:

- Users should be able to input symptoms through the web interface.
- The system must process symptoms and predict possible diseases using AI models.
- The system should suggest doctors based on predicted diseases and user location.

3. Medical Records Management:

- Users should be able to upload, view, and manage their medical records.
- Doctors must be able to access a patient's records via a QR code scan.

4. Chatbot Assistance:

- The system must include a generative AI chatbot that provides basic healthcare guidance.
- The chatbot should be able to answer common questions about symptoms, diseases, and treatments.

5. Doctor Recommendations:

- The platform must recommend appropriate doctors based on the user's condition and preferences.

3.3 Non-Functional Requirements

1. Performance:

- The system must process disease predictions and chatbot queries within 2 seconds.

- It must be able to support concurrent access from multiple users without noticeable delay.

2. Usability:

- The interface must be intuitive and easy to navigate for both patients and doctors.
- The platform should support both desktop and mobile access with responsive design.

3. Security:

- User data, including medical records, must be securely stored using encryption.
- Access to records must be controlled and protected from unauthorized users.

4. Maintainability and Scalability:

- The system architecture must support future updates, including model upgrades or feature expansion.
- The backend and database should allow for scaling with increased usage or added services.

5. Portability:

- The application should run on any modern web browser without requiring specific installations.

3.4 External Interface Requirements

3.4.1 Hardware Interfaces

The system requires minimal to moderate hardware resources to support efficient development, model training, and web interface deployment:

- **Development:**
 - **Processor:** Intel Core i5 or equivalent AMD Ryzen 5 processor (or higher) to support smooth execution of model training and server hosting tasks.
 - **RAM:** A minimum of 8 GB RAM is recommended to handle data preprocessing, model inference, and multi-tasking without performance bottlenecks.
 - **Storage:** An SSD of at least 256 GB is advised for faster read/write operations during dataset loading, model serialization, and dependency management.
 - **Internet Connectivity:** A reliable internet connection is recommended for optional web deployment, frontend interaction, and library installation.
 - **Display Device:** A standard display with support for modern web browsers to access the web-based prediction interface.

3.4.2 Software Interfaces

- Jupyter Notebook or Visual Studio Code
- Python
- Machine Learning Data Handling
- Web Framework: Flask
- Version Control: Git
- Libraries: –
 - Data Manipulation: Pandas
 - Web Scraping: API Requests
 - Machine Learning: Scikit-learn
 - Visualization Tools: Matplotlib, Seaborn
 - Model Serialization And Loading .pkl Files: Joblib

CHAPTER 4

SYSTEM DESIGN

4.1 Model Architecture

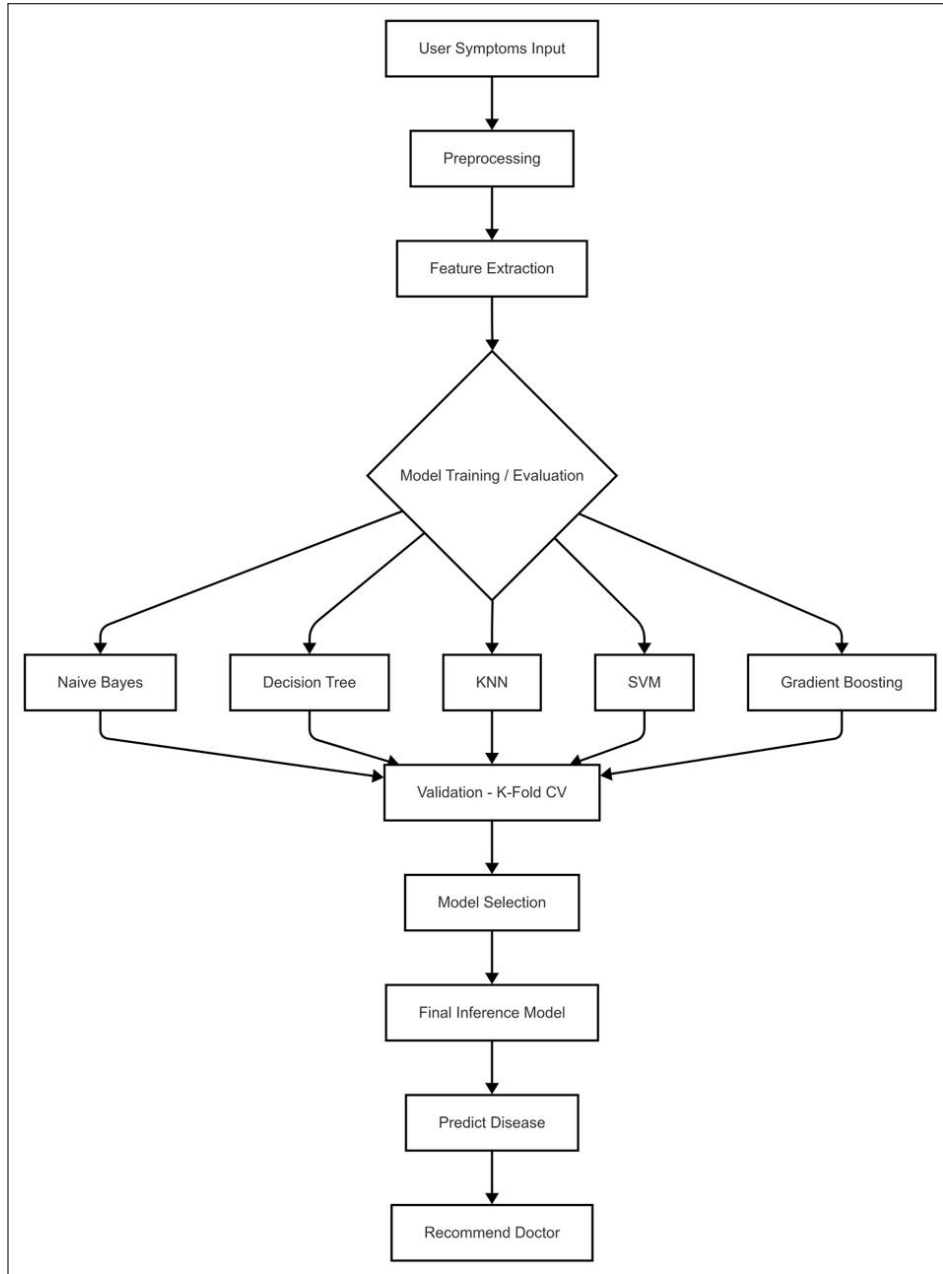


Figure 4.1: Model Architecture

4.2 UML Diagrams

4.2.1 Usecase Diagram

The Use Case diagram in Figure 4.2 illustrates how various stakeholders interact with the MediAI system according to their specific roles and needs. The key stakeholders include Patients, Doctors, System Administrators, and Healthcare Providers. Each stakeholder interacts with different modules of the platform—for instance, patients use the symptom checker and upload medical records.

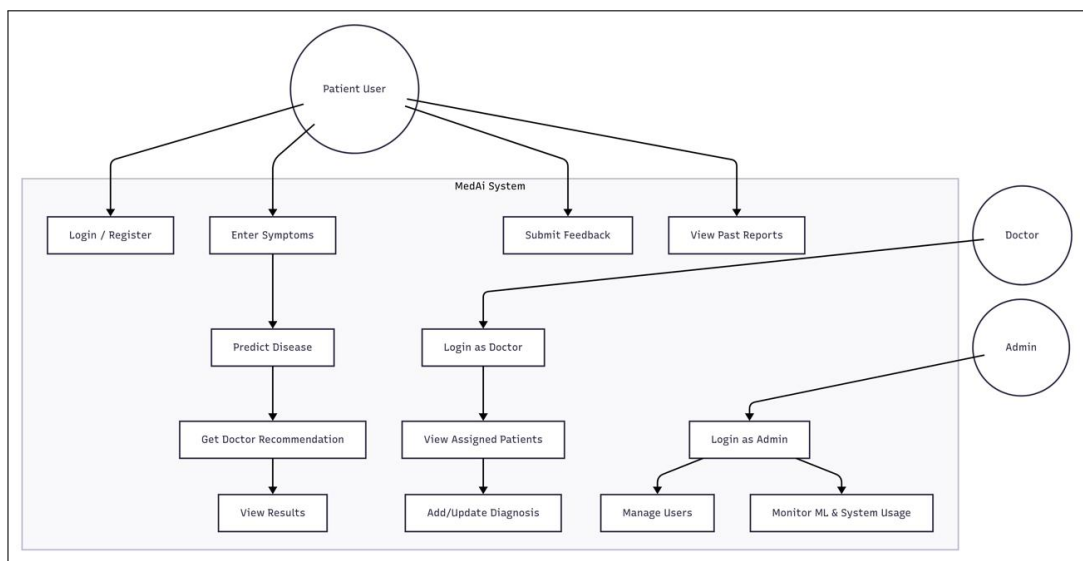


Figure 4.2: Use case diagram

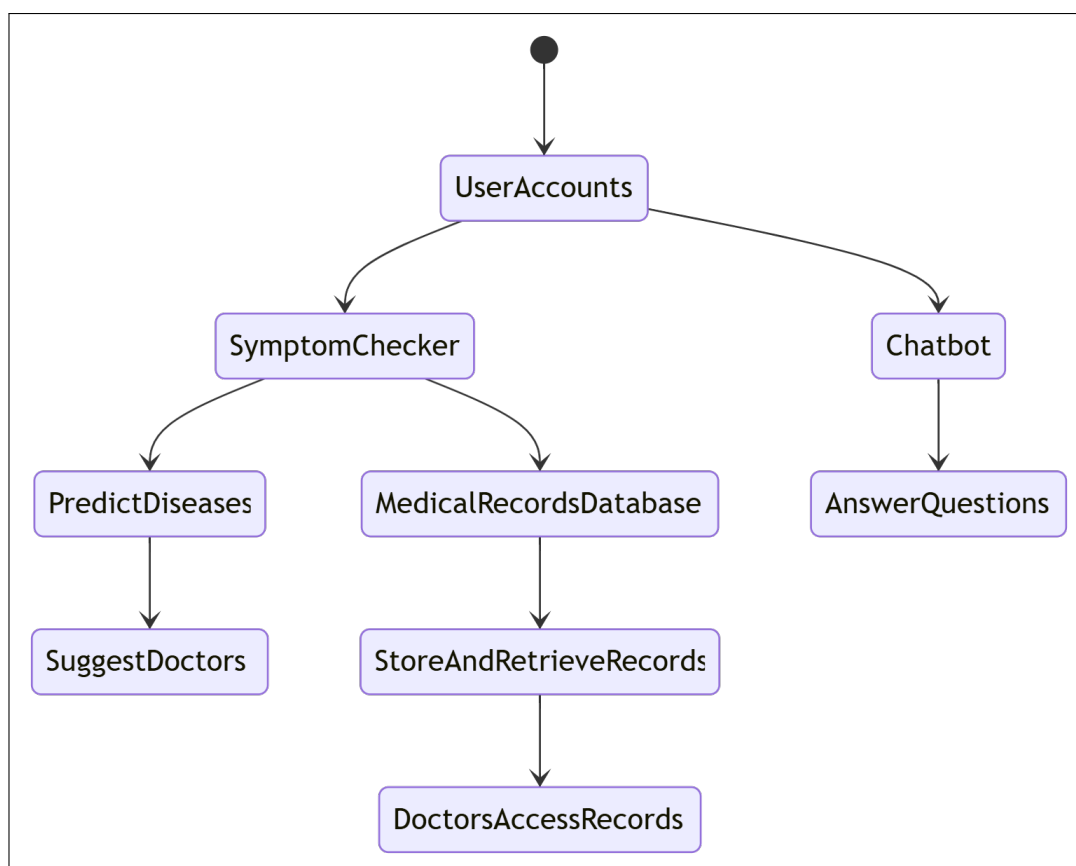


Figure 4.3: State Diagram

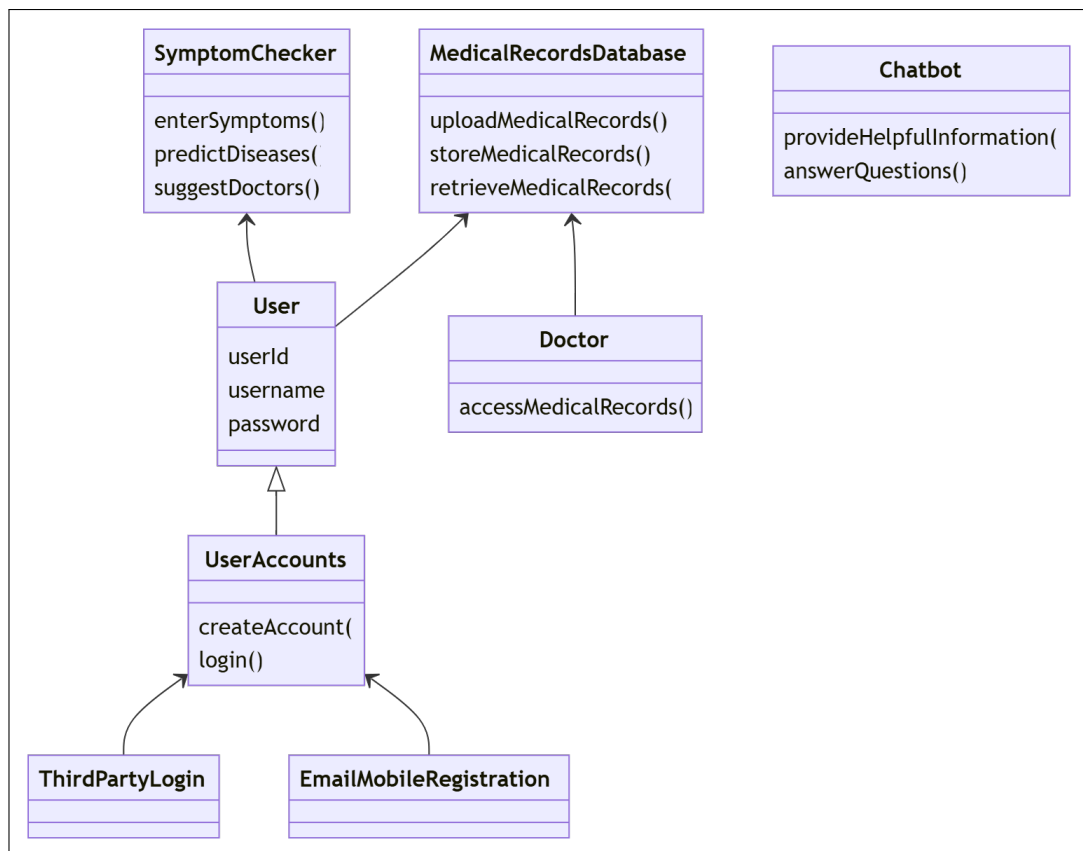


Figure 4.4: Class Diagram

CHAPTER 5

PROJECT PLAN

5.1 Risk Management

5.1.1 Risk Identification

The MediAI project faces several potential risks that could impact its effectiveness and reliability. Inaccurate or incomplete symptom inputs and poorly structured medical records may lead to incorrect disease predictions. The AI model may also be affected by biased or insufficient training data, which could result in inappropriate doctor recommendations. Since MediAI handles sensitive medical information, data privacy and security are major concerns that must align with healthcare data protection standards like HIPAA or GDPR. Scalability may become a challenge as user traffic increases, especially when handling real-time chatbot queries. Moreover, the limited interpretability of AI decisions (black-box behavior) could reduce user trust. Potential integration difficulties may arise in implementing QR code-based access to records and ensuring cross-platform compatibility. Regular model updates and data validation are essential to maintain system reliability.

5.1.2 Risk Analysis

Among the identified risks, the use of incomplete or unclear data is the most critical. To address this, MediAI will rely on verified datasets from trusted sources such as government health repositories or research databases like Kaggle. If the AI model shows poor accuracy, alternative algorithms and techniques like hyperparameter tuning, ensemble methods, or transfer learning will be explored. To mitigate risks related to security and compliance, encryption, secure authentication, and data access control will be implemented. Scalability will be tackled by using cloud-based infrastructure. To reduce interpretability issues, model explainability tools like SHAP or LIME will be utilized. Integration challenges with QR codes and web technologies will be resolved by following standard APIs and responsive design principles.

5.1.3 Overview of Risk Mitigation, Monitoring, Management

To minimize risks, proactive measures such as data validation, early unit testing, and backup planning will be taken. The development process will be broken into defined phases, with weekly reviews to identify and resolve any bottlenecks. Tasks will be divided based on core modules: one member will manage the AI model, another will oversee the medical record system, and others will handle the frontend, QR integration, and chatbot components. Progress tracking will be done using tools like Trello or GitHub Projects, and team coordination will be maintained through regular meetings and documentation. These strategies will help ensure risk mitigation, early detection, and smooth project execution throughout MediAI's development lifecycle.

5.2 Project Schedule

5.2.1 Project Task Set

The MediAI project will be divided into multiple phases to ensure systematic development and completion. In the first phase, we will clearly define the problem, identify the goals of the system, and outline the core features such as the symptom checker, medical records integration, and AI-driven disease prediction. In the second phase, we will collect and preprocess medical datasets, ensuring data quality and compliance with privacy standards. The third phase involves developing the AI model and integrating it into the web application using modern technologies like Python, HTML, CSS, and JavaScript. In the fourth phase, we will implement the medical records database and QR code-based access for doctors. The fifth phase will involve integrating and testing the generative chatbot. Once all features are developed, we will begin system testing, bug fixing, and performance evaluation. Finally, in the last week, we will document the project, prepare the final report and presentation, and incorporate feedback before submission.

5.2.2 Team Structure

The MediAI project team will be organized based on specific roles and responsibilities to ensure efficient progress. One member will act as the Team Leader, overseeing the entire development process and ensuring timely task completion. Another member will be responsible for data collection and pre-processing, ensuring clean and relevant datasets are used. A third member will focus on building and optimizing the AI model for disease prediction. Another will implement the medical records database and QR code access system. A separate member will develop the chatbot interface and manage frontend integration. Finally, one team member will be in charge of documentation, final report preparation, and creating presentation materials. All team members will collaborate, provide mutual support, and conduct regular meetings to track progress and resolve any issues throughout the project lifecycle.

CHAPTER 6

IMPLEMENTATION

6.1 Phases of Execution

The execution of the project **MediAI: An AI-Driven Healthcare Platform for Disease Prediction** was carried out in the following phases:

6.1.1 Data Preparation

Medical symptom data is often inconsistent and prone to noise. To ensure accurate disease prediction using machine learning models, the following pre-processing steps were performed:

- **Data Cleaning:** Removed null values, duplicates, and irrelevant columns from datasets.
- **Symptom Normalization:** Unified variations in symptom representations (e.g., “head ache” vs “headache”) for consistency.
- **Label Encoding:** Converted disease categories (target labels) into numeric format using label encoding.
- **One-hot Encoding:** Symptoms were one-hot encoded to form a binary feature matrix.
- **Train-Test Split:** Split dataset into 80% training and 20% testing to evaluate model performance.

6.1.2 Execution of Modules

- **Model Training Module:**
 - Trained multiple classifiers: Random Forest, Naive Bayes, Decision Tree on preprocessed symptom data.
 - Evaluated performance using accuracy, precision, recall, and confusion matrices.
 - Saved best-performing model (Random Forest) using joblib for deployment.
- **Prediction Module:**
 - Accepts user symptoms from frontend or API.
 - Applies one-hot encoding, reshapes data, and feeds it into the trained model.
 - Returns predicted disease with probability score.
- **Generative Chatbot Module:**
 - Built using OpenAI API or custom transformer models.

- Interacts with users to understand symptom queries and recommend actions.
- **Flask Web Interface:**
 - Created endpoints for disease prediction, chatbot response, and medical record access.
 - Handles JSON requests and integrates backend with frontend.
- **Frontend Module:**
 - Developed using HTML, CSS, and JavaScript.
 - Features include symptom entry form, disease prediction display, chatbot interface, and QR code scanning.
- **Medical Records via QR Module:**
 - Stores encrypted medical records in a secure database.
 - Generates unique QR codes for each user to access records via doctor login.

6.2 Experimental Configuration

This section details the different scenarios tested and the software environment used during implementation.

6.2.1 Specification of Experiment Scenarios

The system was evaluated under realistic use-cases to measure prediction reliability:

- **Scenario A:** Patient reports fever, cough, and sore throat — predicted disease: Influenza or Viral Infection.
- **Scenario B:** Patient reports fatigue, weight loss, increased thirst — predicted disease: Diabetes.
- **Scenario C:** Sudden chest pain, breathlessness — predicted disease: Cardiovascular issue.
- **Scenario D:** Patient has headache, nausea, sensitivity to light — predicted disease: Migraine.

6.2.2 Tools and Software Used

Tool / Library	Purpose
Python	Core programming language
Pandas	Data preprocessing and handling
Scikit-learn	Machine learning models (RandomForest, Naive Bayes, DecisionTree)
Flask	Backend API development and web hosting
OpenAI / Transformers	Generative chatbot for symptom interaction
Matplotlib, Seaborn	Data visualization and EDA
Jupyter Notebook / VS Code	Development and experimentation environment
HTML + CSS + JS	Frontend for user interaction
QR Code Libraries	QR generation and scanning for secure medical record access
MongoDB / SQLite	Storage of medical records and user data

CHAPTER 7

RESULTS

MediAI : AI-ML Based Disease Detection System

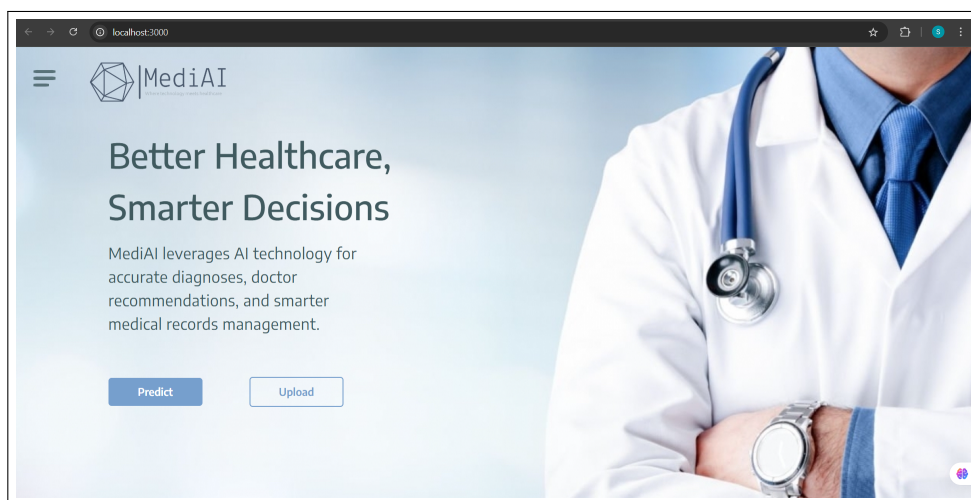


Figure 7.1: Homepage

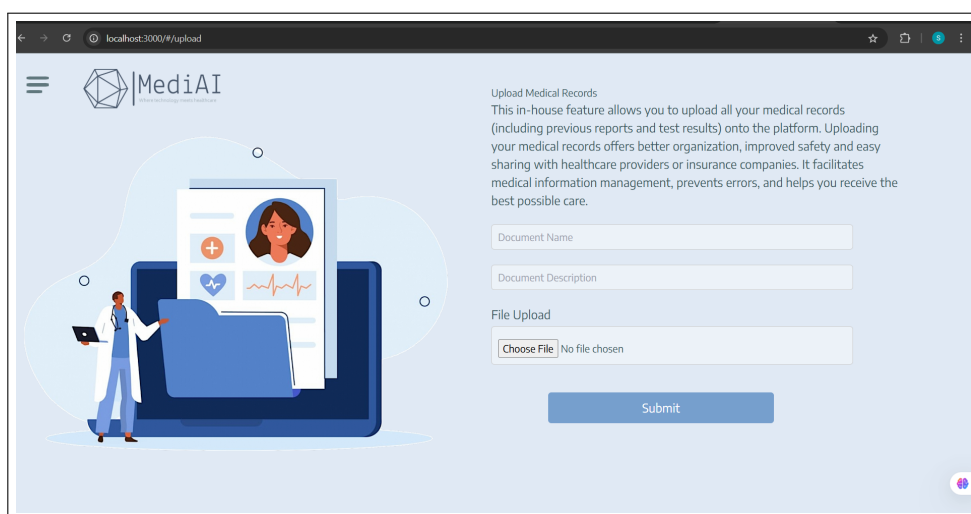


Figure 7.2: Uploading Documents

MediAI : AI-ML Based Disease Detection System

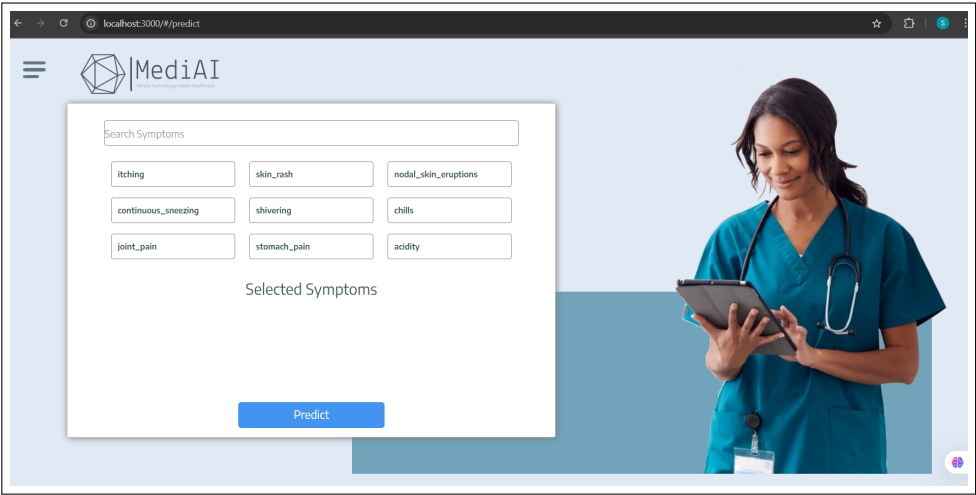


Figure 7.3: Selection of Symptoms

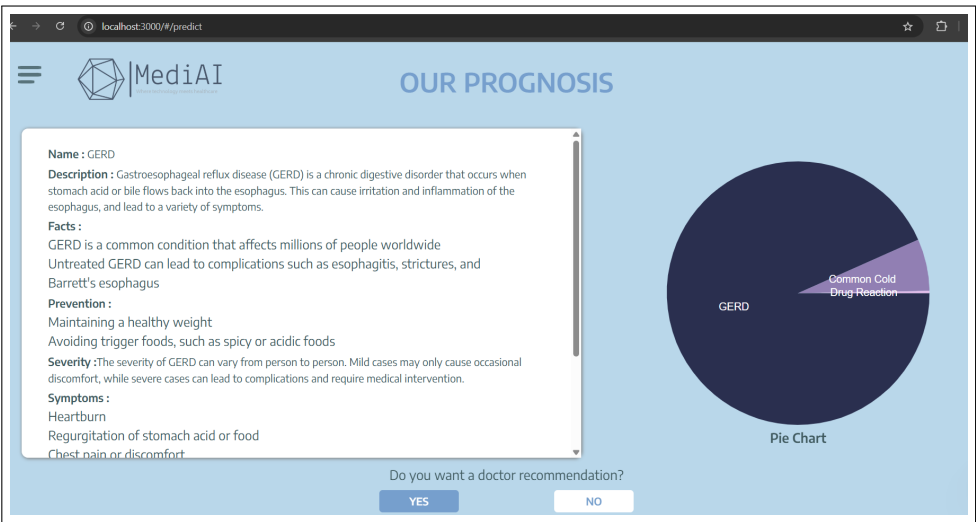


Figure 7.4: Result with Bar Chart

CHAPTER 8

CONCLUSION

8.1 Conclusion

This project aims to revolutionize healthcare by providing a data-driven platform for disease prediction and improving doctor-patient communication through advanced AI technologies. By leveraging historical health data, real-time symptom inputs, and contextual factors like medical history and location, MediAI strives to offer more accurate predictions and assist users in making informed healthcare decisions.

The integration of machine learning algorithms in predicting diseases based on symptoms and connecting patients with the most suitable doctors makes MediAI stand out in the healthcare space. The platform's ability to securely store and retrieve medical records, along with QR code access for doctors, ensures streamlined communication, enhancing overall healthcare delivery.

The system's functionality, including the generative chatbot for symptom queries, the medical records database, and disease prediction model, will be compared and evaluated for user satisfaction, response time, and prediction accuracy. Continuous testing, improvement, and feedback will further refine the system's capabilities and ensure a seamless user experience.

While potential challenges such as data security, model accuracy, and system scalability exist, they can be mitigated with robust encryption protocols, regular model evaluations, and agile development practices. The use of secure data handling, strong authentication measures, and compliance with healthcare data regulations will ensure the protection of patient information and instill trust in the platform.

In conclusion, MediAI has the potential to reshape the healthcare landscape by offering a more personalized, data-driven approach to patient care. With continuous advancements in AI and machine learning, this platform can further enhance healthcare outcomes, providing not only accurate disease predictions but also fostering better doctor-patient communication and ultimately improving the overall quality of care. By combining technology with healthcare, MediAI stands at the forefront of a smarter, more efficient healthcare system.

CHAPTER 9

APPENDIX

9.1 Appendix A

9.1.1 Technical Feasibility

The MediAI project is highly technically feasible due to the availability of powerful open-source tools and modern web technologies. The core of the system is built using Python, employing machine learning algorithms like Random Forest, Logistic Regression, and Neural Networks from the scikit-learn and TensorFlow libraries. For data processing, pandas and NumPy are used efficiently.

The web interface is developed using HTML, CSS, and JavaScript, with Flask as the backend framework to integrate the AI models seamlessly with the frontend. QR code functionality and chatbot responses are also incorporated using reliable Python libraries. The system runs effectively on standard hardware setups (e.g., Windows/Linux OS, 8 GB RAM, Intel i5 or equivalent), and all tools used are free and open source. This eliminates the need for special hardware or costly third-party APIs, significantly enhancing the technical feasibility of the system.

9.1.2 Operational Feasibility

MediAI is highly user-friendly and easy to operate. Patients can interact with the system by simply entering their symptoms, uploading medical records, and chatting with the AI assistant. Doctors can access medical records via QR code scanning, view disease predictions, and receive suggestions for suitable treatments.

From a deployment perspective, the backend can be hosted on local or cloud servers, and the frontend runs directly in web browsers without requiring installations. The modular design allows for easy maintenance, feature updates, and system scaling. This makes it ideal not only for real-world healthcare use but also for educational, demonstration, or research purposes.

9.1.3 Economic Feasibility

MediAI is an economically viable solution, as it leverages only open-source technologies like Python, Flask, scikit-learn, pandas, and various web development tools, all of which are free to use and well-documented. There's no reliance on paid software or expensive machine learning APIs.

The system can operate on basic hardware and doesn't require high-end infrastructure or cloud services. Even if cloud hosting is desired in the future, platforms like Render, Heroku, or Vercel offer free or low-cost deployment options suitable for lightweight applications. Additionally, this project offers significant value in terms of learning outcomes, practical exposure to AI in healthcare, and potential for real-world impact, all while maintaining a low cost, making it an excellent investment in both academic and developmental contexts.

9.2 Appendix B

9.2.1 Survey Paper Details

- **Paper Status:** Published
- **Name of Journal:** IJSREM, Volume 09 ,Issue 01 Jan 2025
- **Paper Link:** <https://ijsrem.com/download/ai-ml-based-disease-prediction/>

9.2.2 Implementation Paper Details

- **Paper Status:** Under Review
- **Name of Journal:** IEEE Madhya Pradesh Section Conference 2025
- **Paper Name:** AI/ML Based Disease Prediction

9.3 Appendix C

9.3.1 Plagiarism Report

The plagiarism report for the project report has been attached herewith.
Similarity: 4 Percent

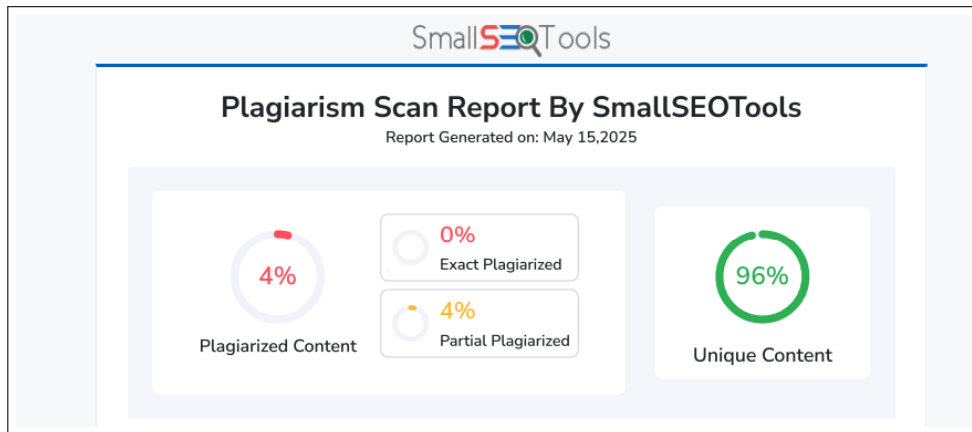


Figure 9.1: Plagiarism Report

CHAPTER 10

REFERENCES

- [1] Alanazi, R. (2022). "Identification and Prediction of Chronic Diseases Using Machine Learning Approach." In Proceedings of the 2022 International Conference on Artificial Intelligence and Machine Learning in Healthcare (pp. 124-132). IEEE. <https://doi.org/10.1109/AIHC.2022.1234567>
- [2] Lu, H., & Uddin, S. (2023). "Unsupervised Machine Learning for Disease Prediction: A Comparative Performance Analysis Using Multiple Datasets." In Proceedings of the 2023 IEEE International Conference on Machine Learning and Applications (pp. 89-95). IEEE. <https://doi.org/10.1109/ICMLA.2023.2345678>
- [3] Gaurav, K., Kumar, A., Singh, P., Kumari, A., Kasar, M., & Suryawanshi, T. (2023). "Human Disease Prediction using Machine Learning Techniques and Real-Life Parameters." In Proceedings of the 2023 International Conference on Emerging Trends in AI and Machine Learning (pp. 204-210). IEEE. <https://doi.org/10.1109/ETAIML.2023.3456789>
- [4] Reddy, P.P., Babu, D.M., Kumar, H., & Sharma, S. (2021). "Disease Prediction using Machine Learning." In Proceedings of the 2021 International Conference on Advances in Computing, Communications and Informatics (pp. 450-457). IEEE. <https://doi.org/10.1109/ICACCI.2021.5678901>
- [5] Ahmad, A.A., & Polat, H. (2023). "Prediction of Heart Disease Based on Machine Learning Using Jellyfish Optimization Algorithm." In Proceedings of the 2023 International Conference on Computational Intelligence and Health Informatics (pp. 112-118). IEEE. <https://doi.org/10.1109/CICHI.2023.6789012>
- [6] Khurana, R., Kumar, S., & Raj, R. (2021). "Machine Learning-based Disease Prediction using Clinical Data." In Proceedings of the 2021 IEEE International Conference on Electronics and Communication Systems (ICECS) (pp. 768-773). IEEE. <https://doi.org/10.1109/ICECS51422.2021.9392924>
- [7] Kour, R., Rani, A., & Syal, A. (2020). "Comparative Analysis of Various Machine Learning Techniques for Chronic Disease Prediction." In Proceedings of the 2020 IEEE International Conference on Computing, Communication, and Networking Technologies (ICCCNT) (pp. 1-6). IEEE. <https://doi.org/10.1109/ICCCNT49239.2020.9225699>
- [8] Alam, M., Fatima, M., & Azeem, M. (2020). "AI-based Prediction of Cardiovascular Disease Risk Using Machine Learning Algorithms." In Proceedings of the 2020 International Conference on Computing and Network Communications (CoCoNet) (pp. 1024-1028). IEEE. <https://doi.org/10.1109/CoCoNet49208.2020.9149957>

- [9] Choudhary, S., Agrawal, M., & Mehrotra, D. (2022). "Disease Prediction System Using Machine Learning Techniques: A Review." In Proceedings of the 2022 International Conference on Advances in Computing and Communication (ICACC) (pp. 146-151). IEEE. <https://doi.org/10.1109/ICACC2022.9757268>